

Training Report: Active Directory & LDAP

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Learning Resources: Microsoft Documentation, Windows Server Lab, VirtualBox, Hands-on Exercises

Executive Summary

I configured a Windows Server 2022 virtual machine as a Domain Controller named **DC01** and initialized the **lab.local** Active Directory domain. Following static IP assignment and Active Directory Domain Services (AD DS) installation, I set up Organizational Units (OUs), user accounts, and security groups to practice centralized directory management.

To explore directory queries via LDAP, I used **LDP.exe** to establish a connection to the Domain Controller on port 389, performed an LDAP bind, and authenticated using the domain administrator account. I then executed targeted LDAP searches to query users and groups, analyze Distinguished Names (DNs), and inspect core attributes including sAMAccountName, member, and memberOf.

Active Directory

Active Directory (AD) is a proprietary directory service developed by Microsoft for Windows domain networks. It serves as a centralized database and set of services that allows system administrators to manage and secure network resources such as users, computers, printers, and security policies from a single location.

- Core Functions: Identity management, authentication (via Kerberos/NTLM), authorization, domain security, and centralized policy enforcement (Group Policy Objects).

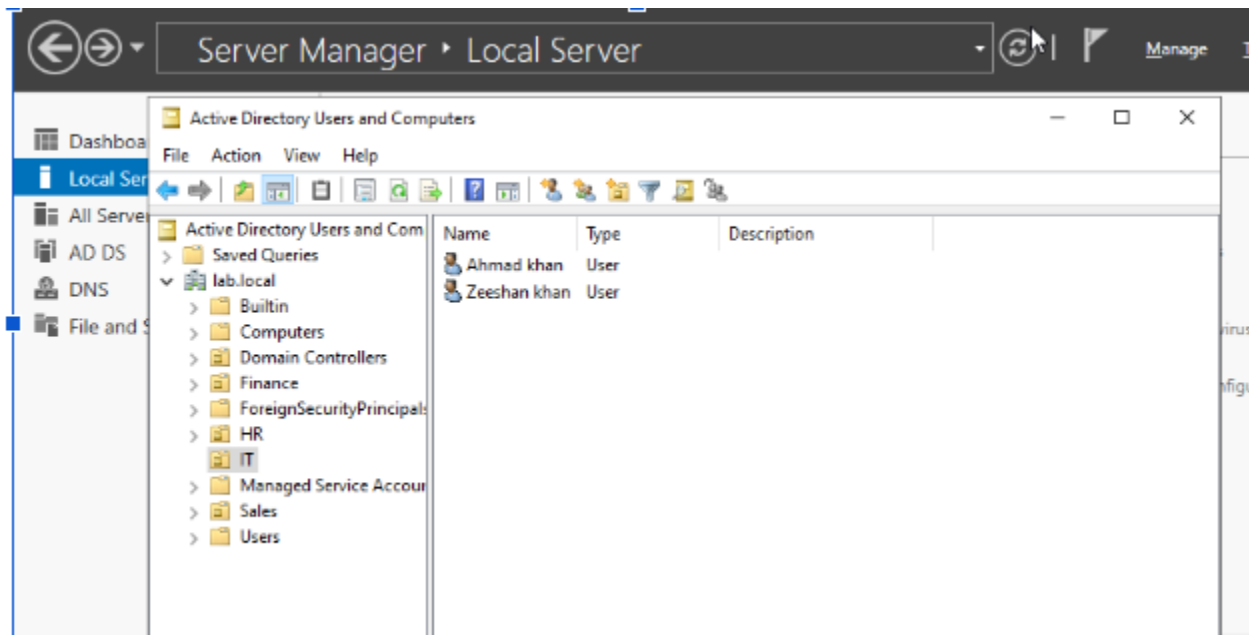
1. Active Directory Setup

- Installed and configured **Windows Server 2022** on VirtualBox.
- Configured a static IPv4 address and promoted the server to a **Domain Controller**.
- Initialized the **lab.local** domain along with integrated **DNS services**.
- Utilized **Active Directory Users and Computers (ADUC)** for administrative management.

2. Directory Structure (OUs)

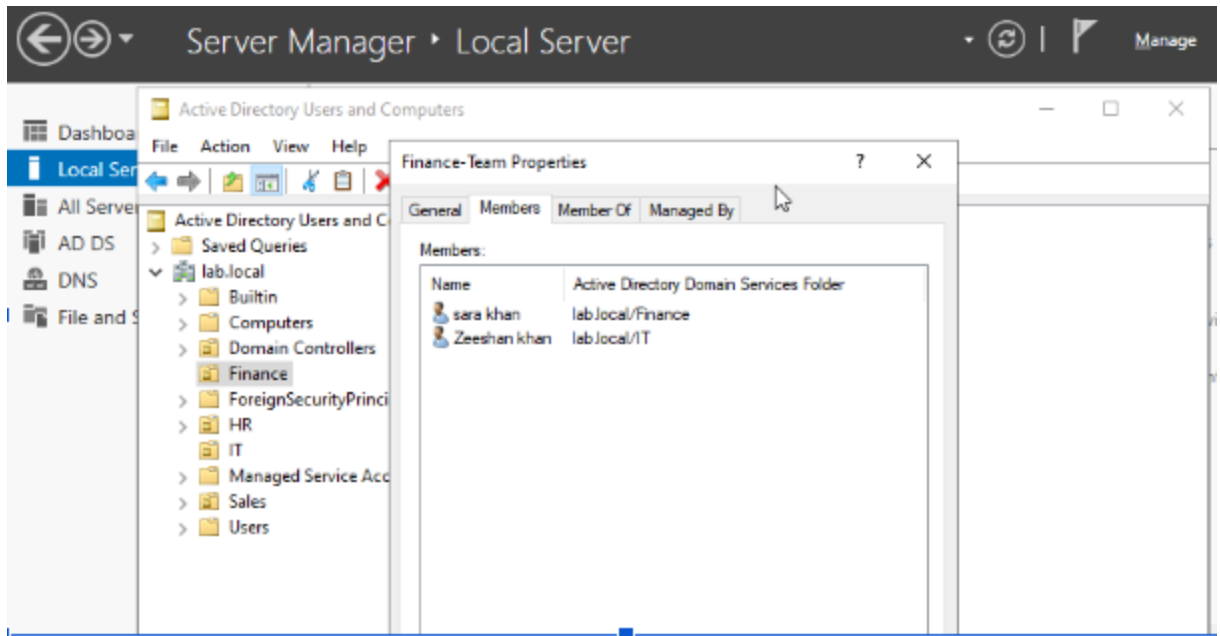
Designed a structured Organizational Unit layout under the root domain:

- **lab.local**
 - **IT**
 - **Finance**
 - **HR**
 - **Sales**



3. User & Group Management

- Provisioned test user accounts across created OUs.
- Created security groups including IT-Team and Finance-Team.
- Assigned users to appropriate groups to evaluate role-based authorization and access management.



LDAP

Lightweight Directory Access Protocol (LDAP) is an open, vendor-neutral application protocol used to query, modify, and manage information stored in a central directory service over an IP network.

- **Core Functions:** Acts as a standard "communication language" that applications use to search for objects, authenticate users, and retrieve group membership details from directory databases like Active Directory or OpenLDAP.

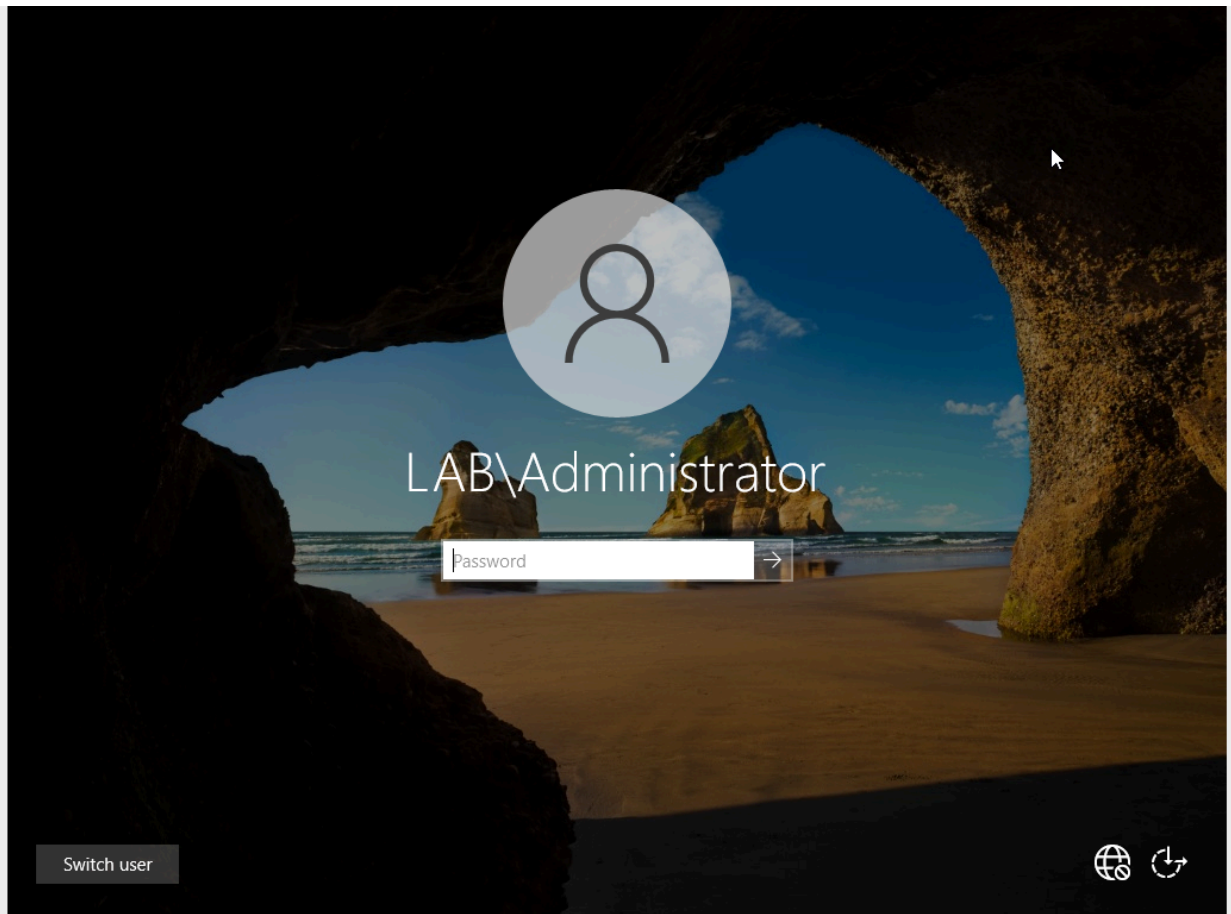
4. LDAP Operations & Inspection

- **Utility:** Used LDP.exe for directory exploration and authentication testing.
- **Connection & Bind:** Connected to DC01 on Port 389 and retrieved **RootDSE** information.
- **Authentication:** Executed an LDAP bind using domain administrator credentials.
- **Queries & Filtering:** Filtered directory objects using LDAP search strings:
 - (objectClass=user) — retrieved all user objects.
 - (sAMAccountName=Administrator) — isolated specific user accounts.
- **Attribute Analysis:** Evaluated object relationship attributes (member and memberOf) and full Distinguished Name (DN) paths.

Lab Environment Configuration

Component	Configuration
Server Name	DC01
Operating System	Windows Server 2022
Domain	lab.local
IP Address	192.168.56.10
Subnet Mask	255.255.255.0
Preferred DNS	192.168.56.10
LDAP Port	389 (Standard Unencrypted)
Network Adapter	VirtualBox Host-Only Adapter

Window Server Installation On VirtualBox



File Machine View Input Devices Help

Server Manager

Server Manager ▸ Dashboard

Network Connections

Administrator: Windows PowerShell

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\Administrator> ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::305d:70c7:d5ec:e277%3
IPv4 Address. : 192.168.56.10
Subnet Mask : 255.255.255.0
Default Gateway :

```
PS C:\Users\Administrator> ping 192.168.56.10
```

Pinging 192.168.56.10 with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.56.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

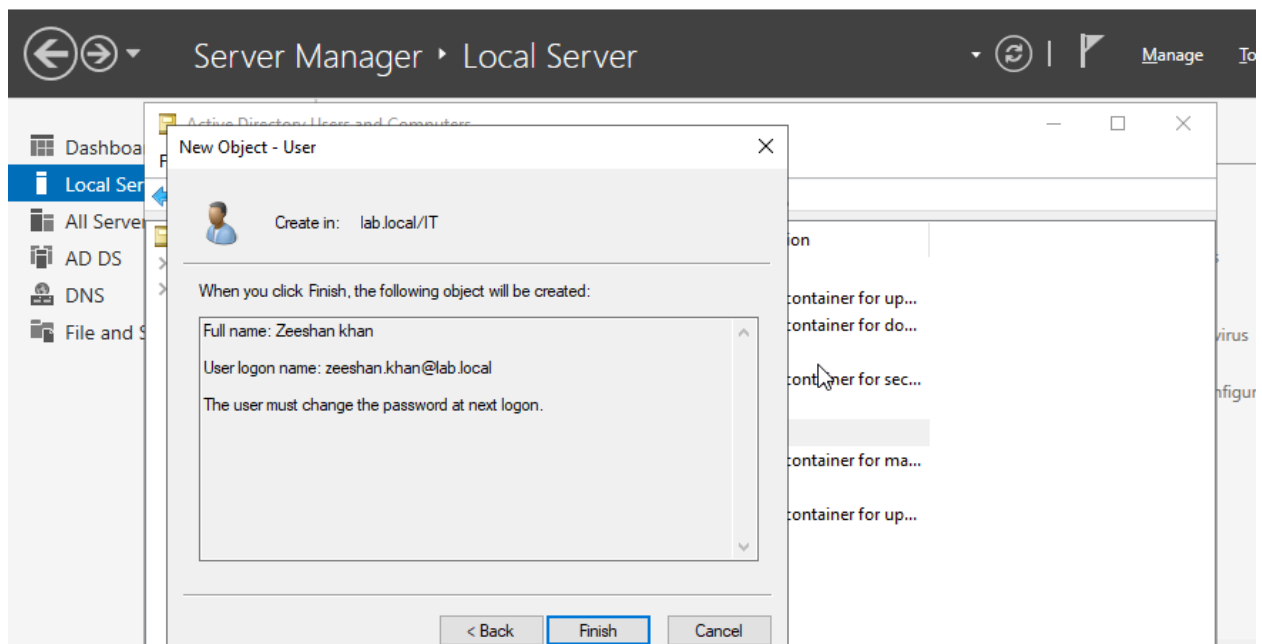
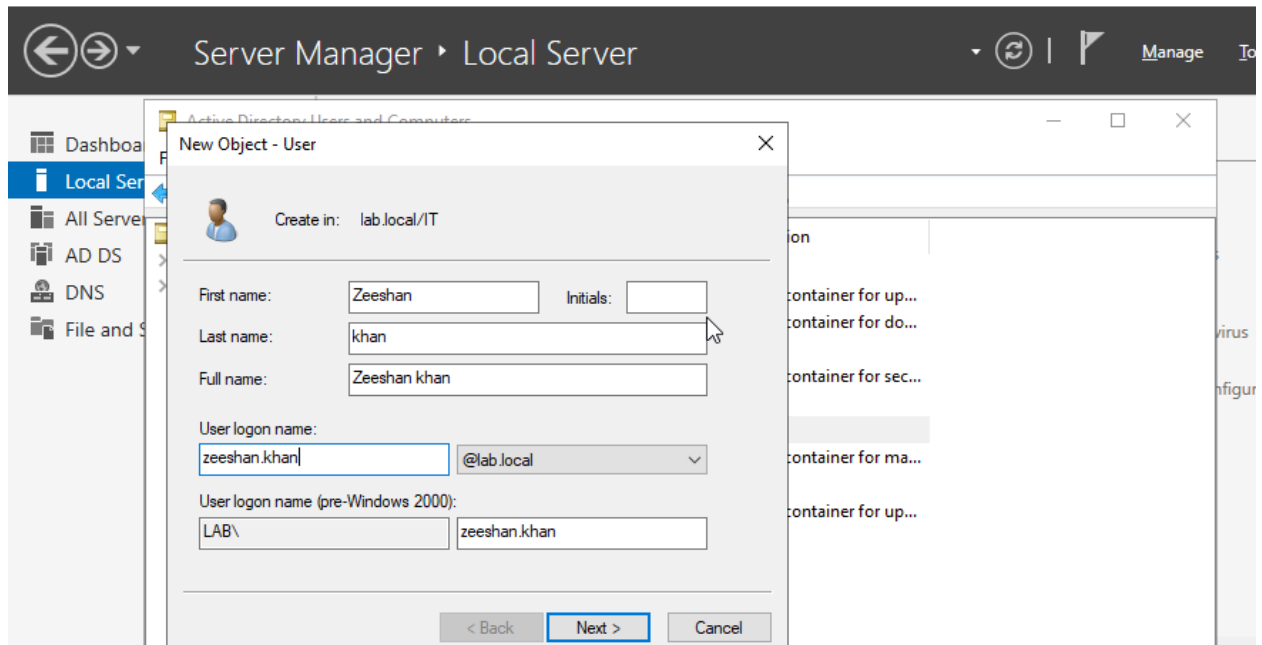
```
PS C:\Users\Administrator>
```

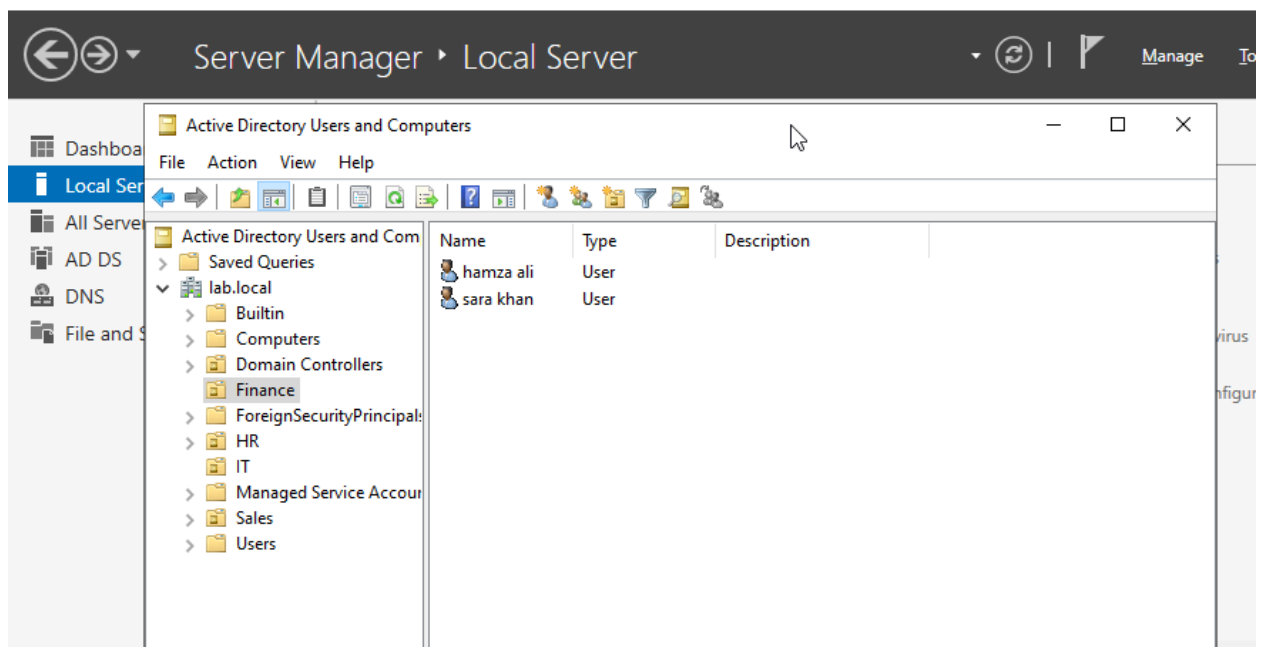
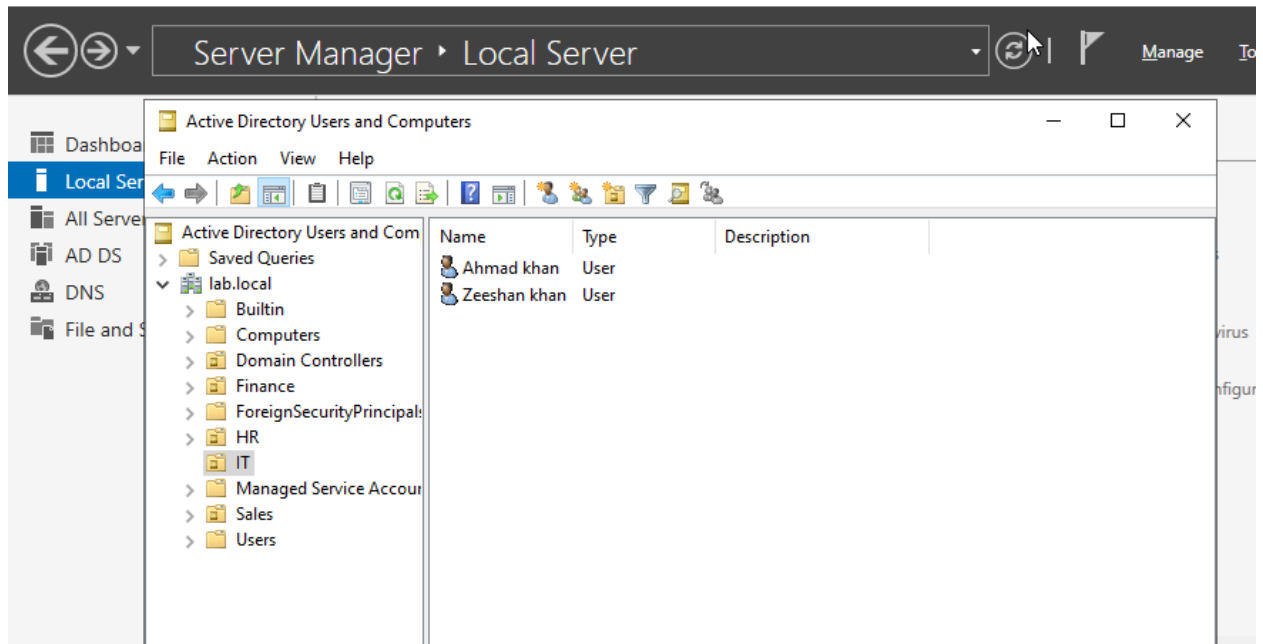
1 item | 1 item selected

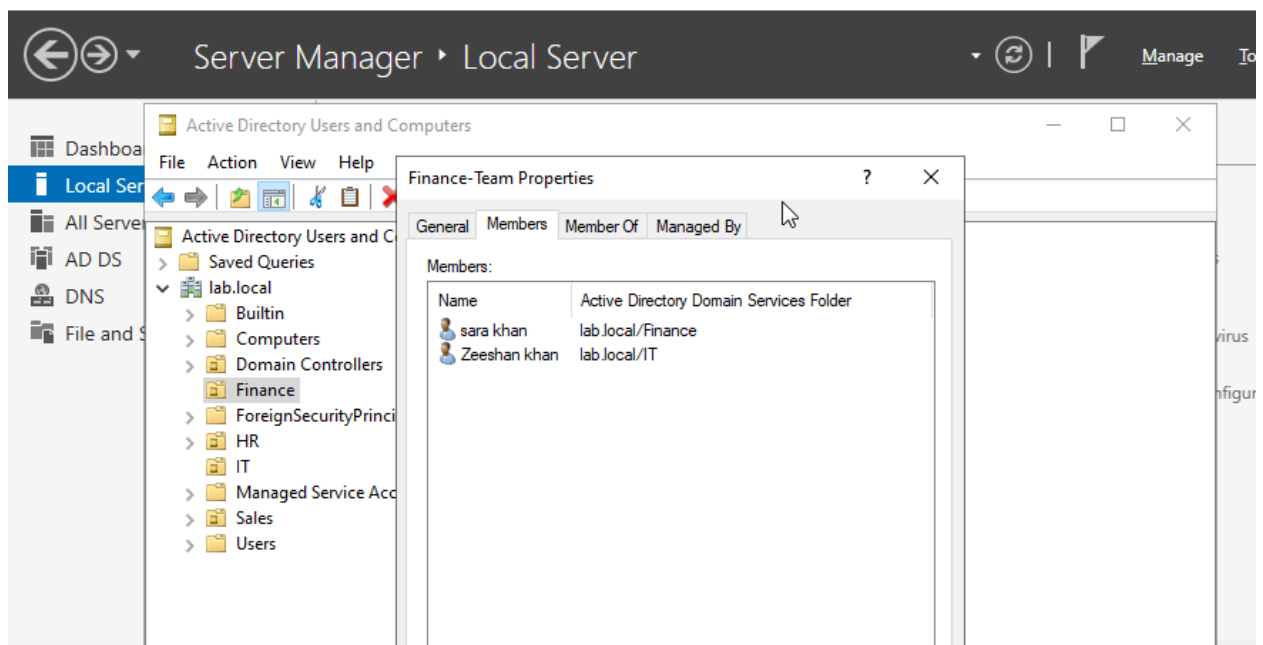
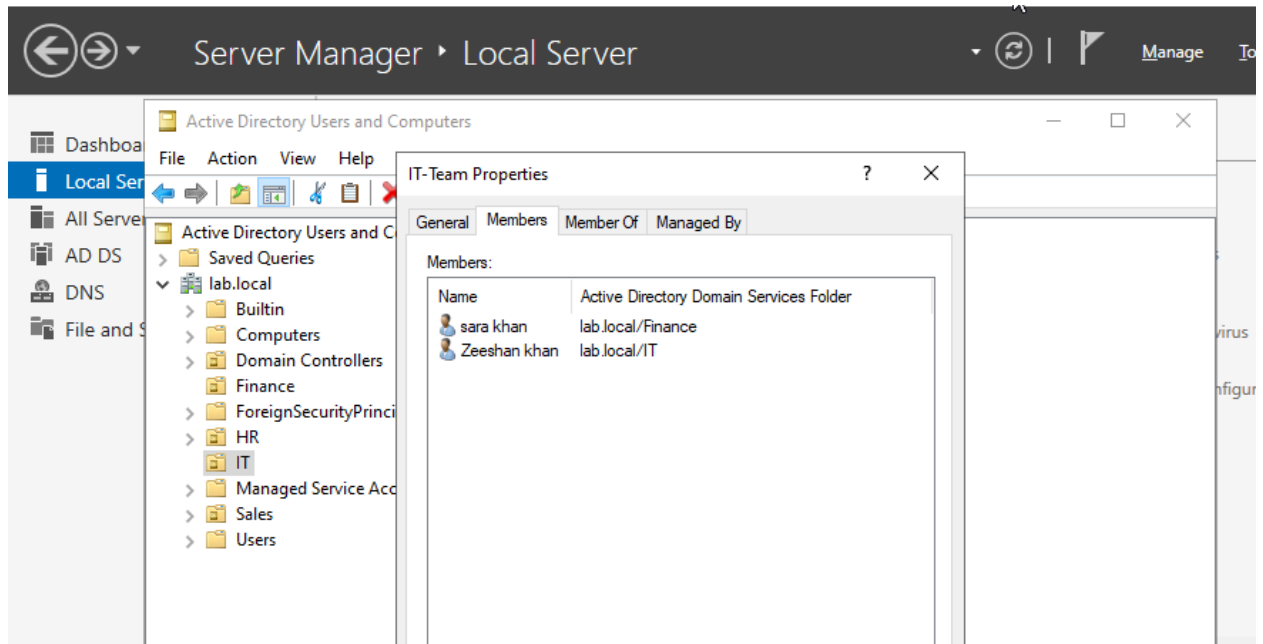
BPA results

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Active Directory Screenshots







```
Administrator: Windows PowerShell

OperationMasterRoles : {SchemaMaster, DomainNamingMaster, PDCEmulator, RIDMaster...}
Partitions            : {DC=ForestDnsZones,DC=lab,DC=local, DC=DomainDnsZones,DC=lab,DC=
CN=Schema,CN=Configuration,DC=lab,DC=local, CN=Configuration,DC=
ServerObjectDN       : CN=DC01,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Config
ServerObjectGuid      : bb27edd0-2bf2-4e0e-aec4-8a319d2810ad
Site                 : Default-First-Site-Name
SslPort              : 636

PS C:\Users\Administrator> Get-ADUser -Filter * | Select Name

Name
----
Administrator
Guest
krbtgt
Zeeshan khan
Ahmad khan
sara khan
hamza ali

PS C:\Users\Administrator> _
```

LDAP Screenshots

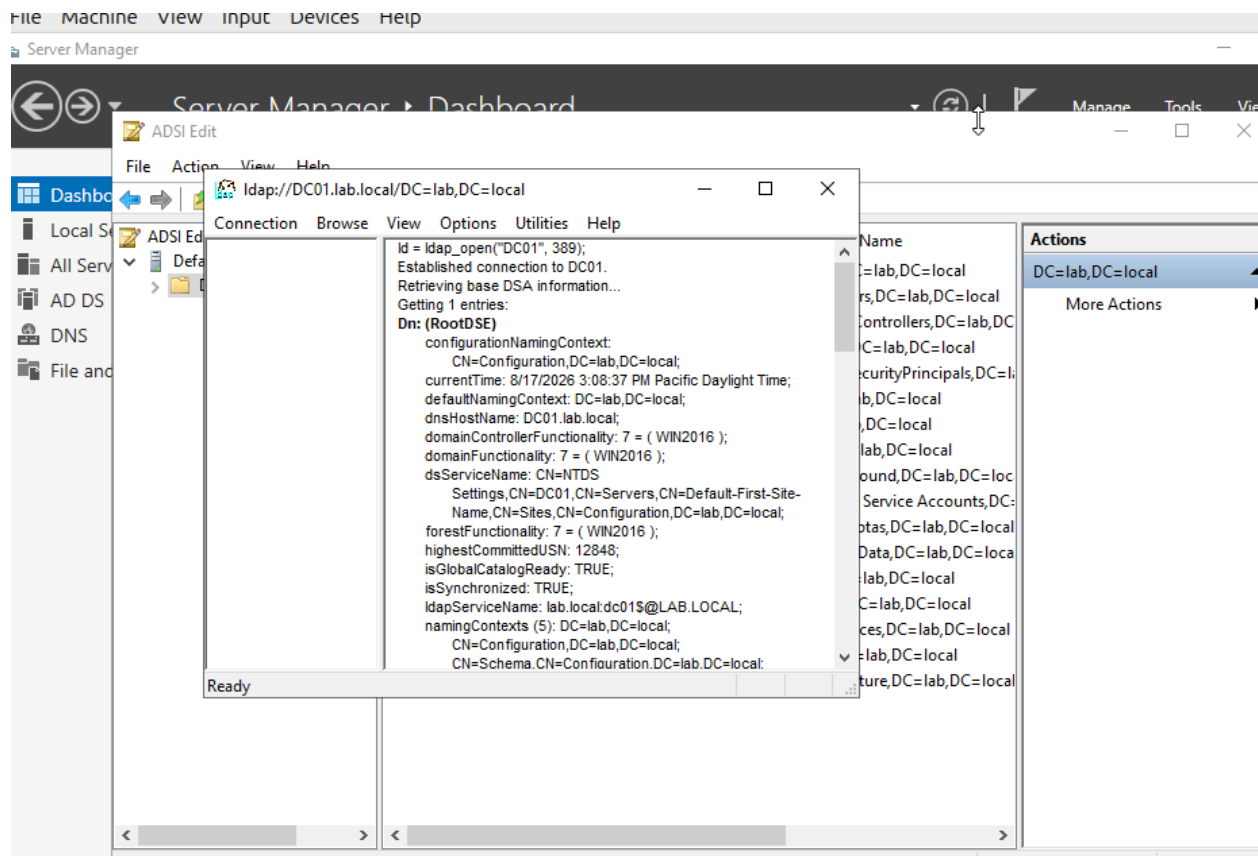
The top screenshot shows the ADSI Edit window. The left pane displays the directory tree structure, with 'DC=lab,DC=local' selected. The right pane shows a table of objects in the directory.

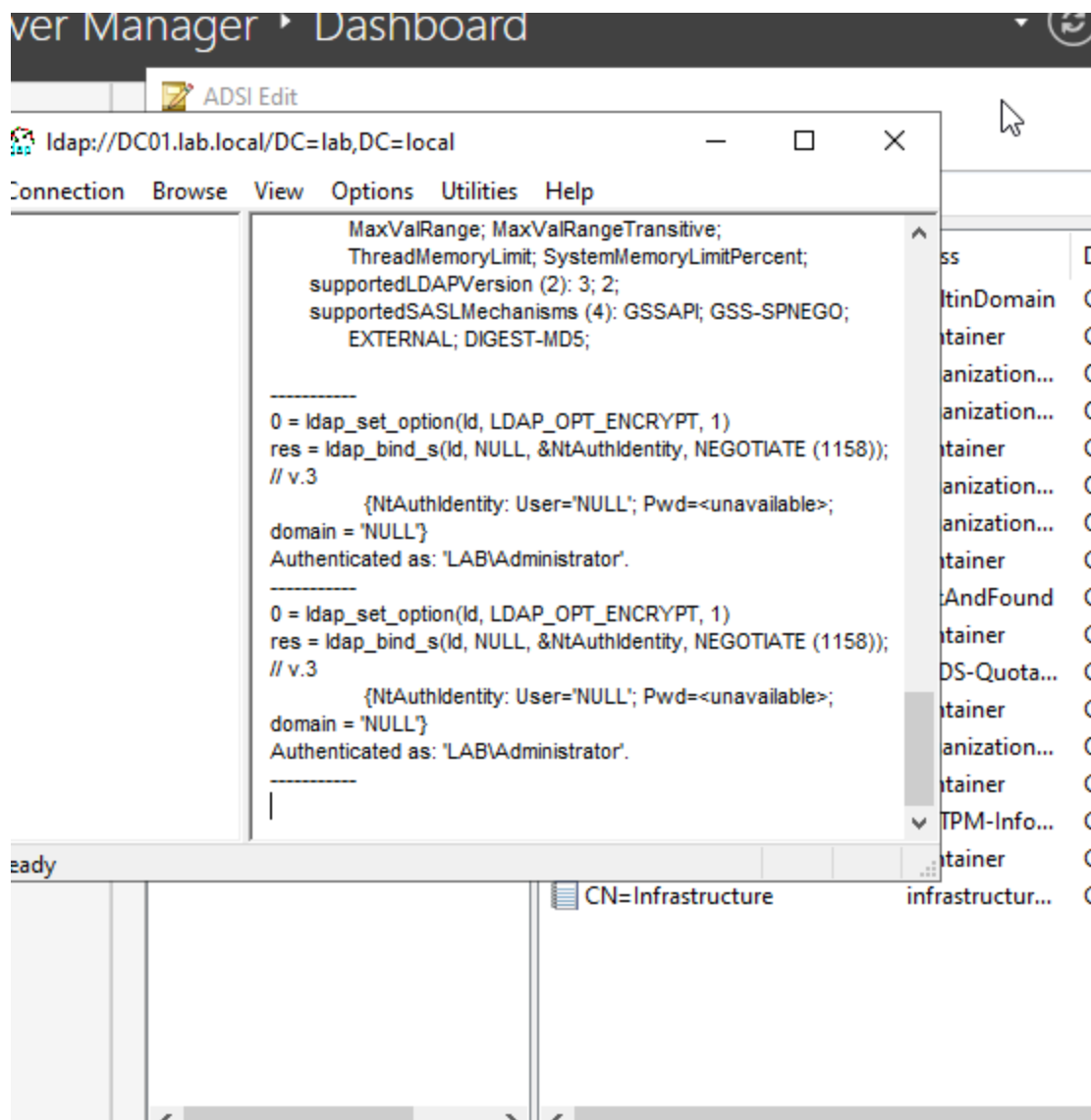
Name	Class	Distinguished Name
CN=Builtin	builtinDomain	CN=Builtin,DC=lab,DC=local
CN=Computers	container	CN=Computers,DC=lab,DC=local
OU=Domain Controllers	organization...	OU=Domain Controllers,DC=lab,DC=local
OU=Finance	organization...	OU=Finance,DC=lab,DC=local
CN=ForeignSecurityPrincip...	container	CN=ForeignSecurityPrincipals,DC=lab,DC=local
OU=HR	organization...	OU=HR,DC=lab,DC=local
OU=IT	organization...	OU=IT,DC=lab,DC=local
CN=Keys	container	CN=Keys,DC=lab,DC=local
CN=LostAndFound	lostAndFound	CN=LostAndFound,DC=lab,DC=local
CN=Managed Service Acco...	container	CN=Managed Service Accounts,DC=lab,DC=local
CN=NTDS Quotas	msDS-Quota...	CN=NTDS Quotas,DC=lab,DC=local
CN=Program Data	container	CN=Program Data,DC=lab,DC=local
OU=Sales	organization...	OU=Sales,DC=lab,DC=local
CN=System	container	CN=System,DC=lab,DC=local
CN=TPM Devices	msTPM-Info...	CN=TPM Devices,DC=lab,DC=local
CN=Users	container	CN=Users,DC=lab,DC=local
CN=Infrastructure	infrastructur...	CN=Infrastructure,DC=lab,DC=local

The bottom screenshot shows the Ldp window. A 'Connect' dialog box is open, displaying the following information:

- Server: DC01
- Port: 389
- ☐ Connectionless
- ☐ SSL

The 'OK' button is highlighted.





LDAP Search Filters:

File Machine View Input Devices Help

Server Manager

Server Manager ▸ Dashboard

Dashboard

- Local Server
- All Servers
- AD DS
- DNS
- File and Storage Services

ADSI Edit

ldaps://DC01.lab.local/DC=lab,DC=local

Connection Browse View Options Utilities Help

***Searching...
ldap_search_s(ld, "DC=lab,DC=local", 2, "(objectclass=user)", attrList, 0, &msg)
Getting 8 entries:
Dn: CN=Administrator,CN=Users,DC=lab,DC=local
canonicalName: lab.local/Users/Administrator;
description: Built-in account for administering the computer/domain;
name: Administrator;
objectClass (4): top; person; organizationalPerson; user;
Dn: CN=Guest,CN=Users,DC=lab,DC=local
canonicalName: lab.local/Users/Guest;
description: Built-in account for guest access to the computer/domain;
name: Guest;
objectClass (4): top; person; organizationalPerson; user;
Dn: CN=DC01,OU=Domain Controllers,DC=lab,DC=local
canonicalName: lab.local/Domain Controllers/DC01;
name: DC01;
objectClass (5): top; person; organizationalPerson; user; computer;

Ready

CN=Infrastructure

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Search

Base DN: DC=lab,DC=local

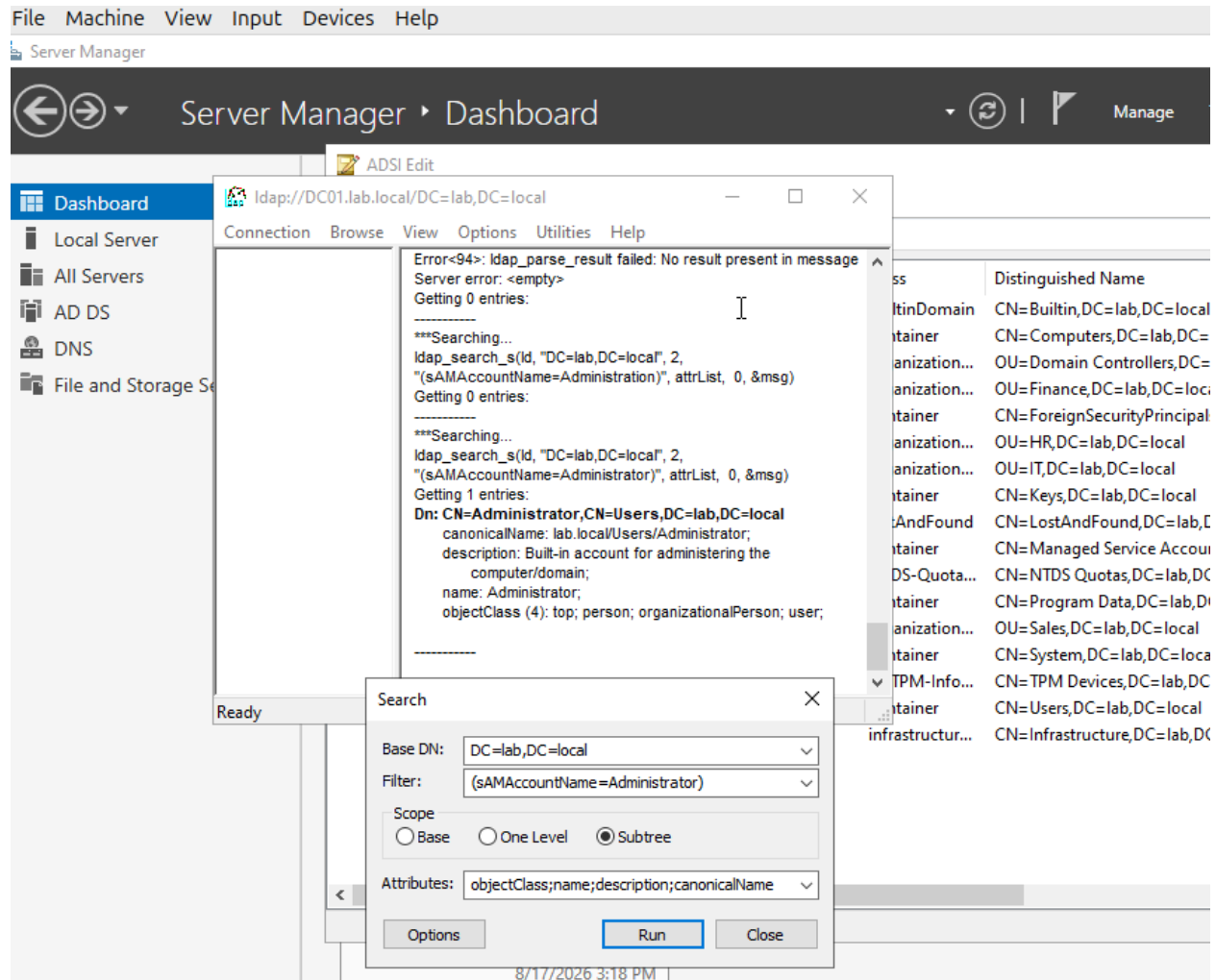
Filter: (objectclass=user)

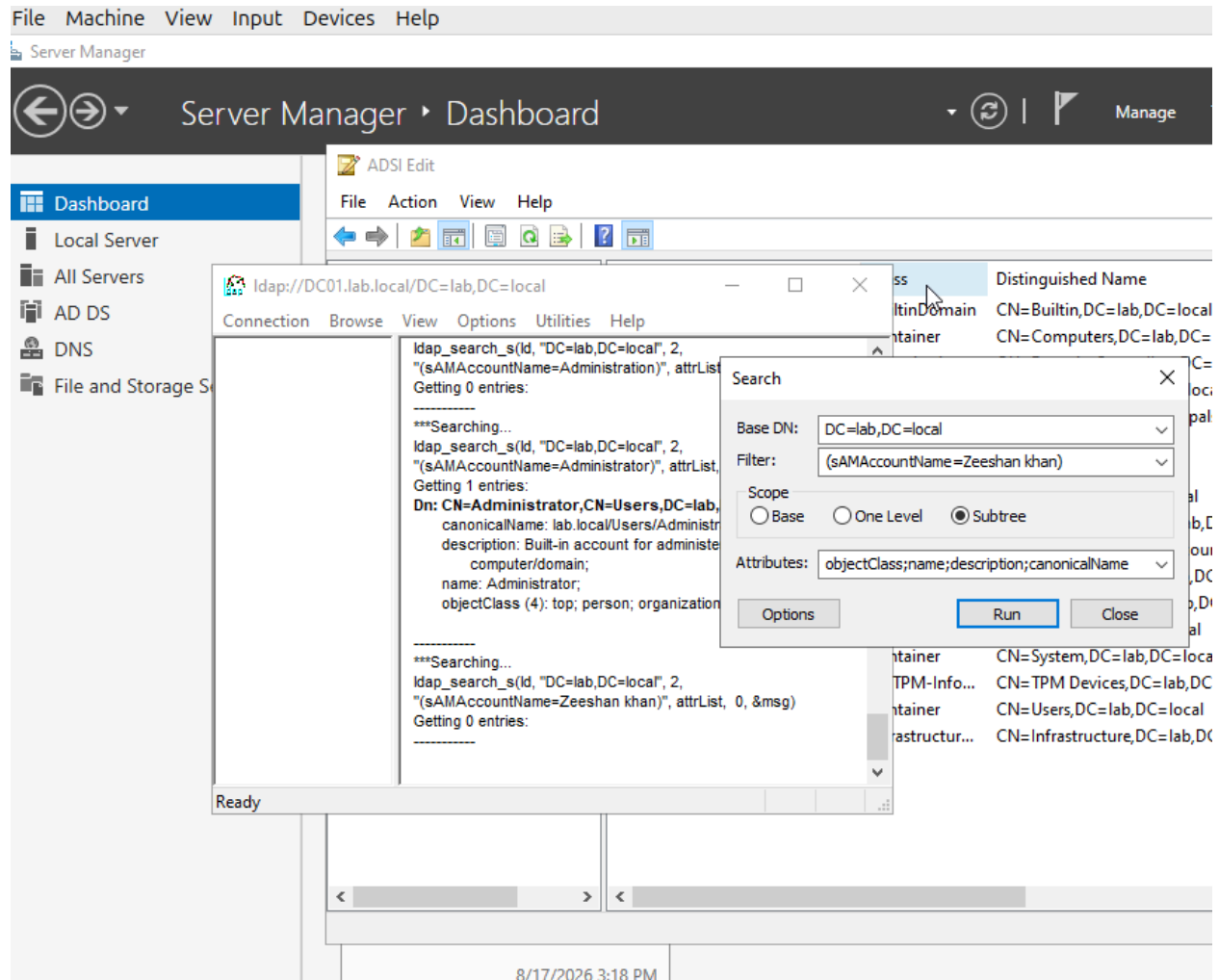
Scope
☐ Base ☐ One Level ☒ Subtree

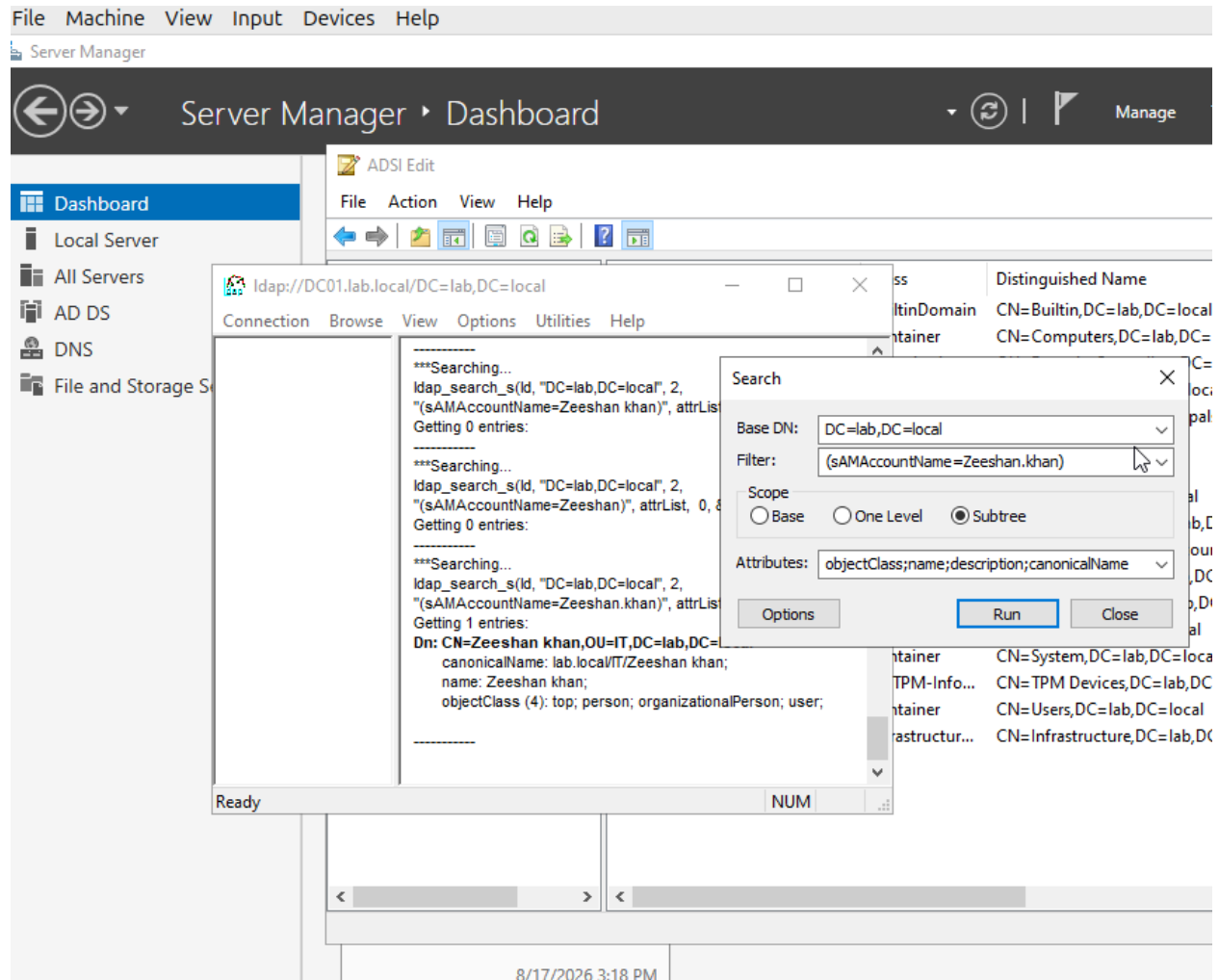
Attributes: objectClass;name;description;canonicalName

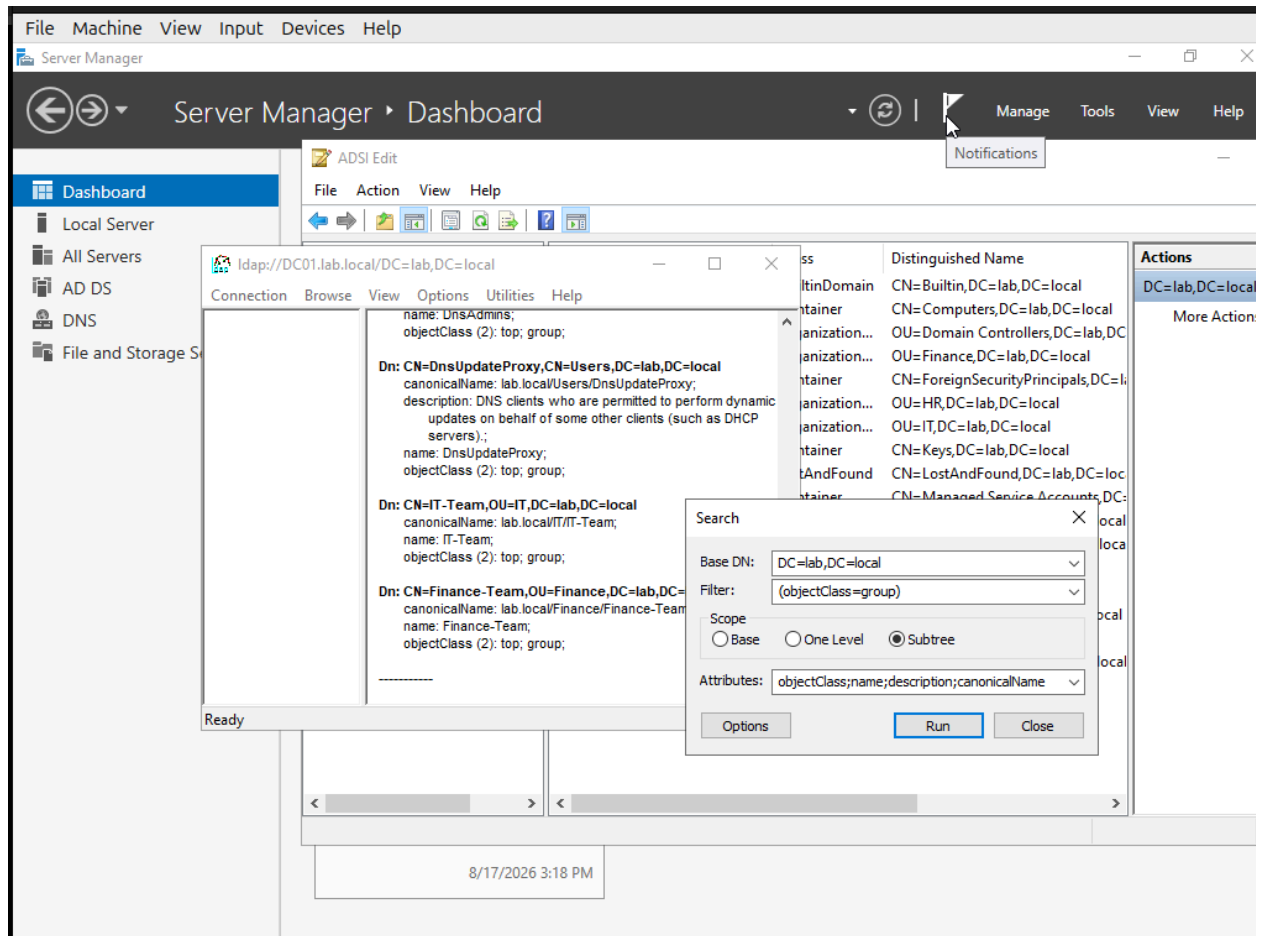
Options Run Close

	Distinguished Name
Administrator	CN=Builtin,DC=lab,DC=local
Guest	CN=Computers,DC=lab,DC=local
Domain Controllers	OU=Domain Controllers,DC=lab,DC=local
Finance	OU=Finance,DC=lab,DC=local
ForeignSecurityPrincipal	CN=ForeignSecurityPrincipal,DC=lab,DC=local
HR	OU=HR,DC=lab,DC=local
IT	OU=IT,DC=lab,DC=local
Keys	CN=Keys,DC=lab,DC=local
LostAndFound	CN=LostAndFound,DC=lab,DC=local
Managed Service Account	CN=Managed Service Account,DC=lab,DC=local









Conclusion

This lab exercise provided hands-on experience in deploying Active Directory infrastructure and interfacing with directory services via LDAP. Key takeaways include mastering domain promotion, designing structured Organizational Units, managing group-based permissions, and directly querying directory attributes using LDAP search filters.